



RETAIL VISION DASHBOARD 2023

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Date: 06/03/2026

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- I declare that where I have used a language translation software in the completion of this assignment I have clearly stated where and why I have used it.
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Signed: Shivaani Raju

Date: 06/03/2026

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1) INTRODUCTION

Data visualisation is a graphical representation which is not only about making charts, but about giving raw numbers a form, that people can genuinely read and interpret. Once data is presented visually, patterns and relationships that would have gone unnoticed in a spreadsheet start to become a lot clearer.

The main idea for this project came from everyday behaviour of consumers. Shopping is a day to day activity that most people can relate to from personal experience, which made it a practical and engaging subject to base the analysis around. The project is centred around a shopping behaviour dataset from various states across the U.S. and the idea was to see how consumer habits and preferences differ across regions and customer groups.

An interactive web application was built using R Shiny so these insights could be explored in a more hands on way. By bringing different visualisation tools together into one web interface, the project was developed into a dashboard that shows how different groups of consumers behave and what their preferences look like. The code for this dashboard was developed by knowledge gained from course materials, tutorials and publicly available visualisation examples. **[Question a]**

2) DATASET

The original dataset has 3,900 individual shopping records and 16 variables. The data was sourced from Kaggle (Customer Shopping Behaviour Analysis by Ward Abilal). Following data pre-processing, a 17th variable “Purchase Date” was synthesised and added to the dataset to support time-series analysis(*Customer Shopping Behaviour Analysis*, no date).

Prior work on this dataset exists on Kaggle, but the visualisations done were at basic levels. This project does not take any reference from those previous analyses, the visualisations and ideas here were developed entirely on my own. The dataset used is a modified version with the synthesised Purchase Date column added, which opens up time-series and seasonal analysis that the original dataset does not support. **[Question b]**

Table 1: Dataset Variable Descriptions

Column Name	Description
Customer ID	Unique customer identifier
Age	Age of the customer
Gender	Gender of the customer
Item Purchased	Name of the product purchased
Category	Product category
Purchase Amount (USD)	Price of the product
Location	Customer's location (U.S. states)
Size	Size of the product
Colour	Colour of the product
Season	Season when the purchase was made
Review Rating	Customer rating of the product
Subscription Status	Whether the customer holds a subscription
Discount Applied	Indicates whether a discount was applied
Previous Purchases	Number of previous purchases by the customer
Payment Method	Mode of payment used
Frequency of Purchases	How often the customer makes purchases
Purchase Date	Date of the purchase (synthesised)

3) DATA PRE-PROCESSING

The original dataset came out with no date column, which ruled out any time- series work until the data was modified. A Purchase Date variable was built using R, and the lubridate library was used to take care of the date formatting across all 3,900 rows. The Season column in each row was used to figure out which months were valid for that specific record. In the case of Winter, only December, January and February were in the pool. A random day was then drawn from within whichever month got selected. Every date sat within 2023, and a seed of 42 was applied for reproducibility. The resulting 17 variable dataset was saved as a CSV and loaded into the R shiny app.

4) VISUALISATIONS

This section goes through the visualisations built during the project, each one tackling a different part of the dataset. Seasonal trends, demographic distributions, geographic revenue and customer behaviour. The aspects of data analysis investigated included pattern recognition, distribution comparison, time-series forecasting, spatial analysis and behavioural analysis. **[Question f]**

4.1) DASHBOARD HOME PAGE

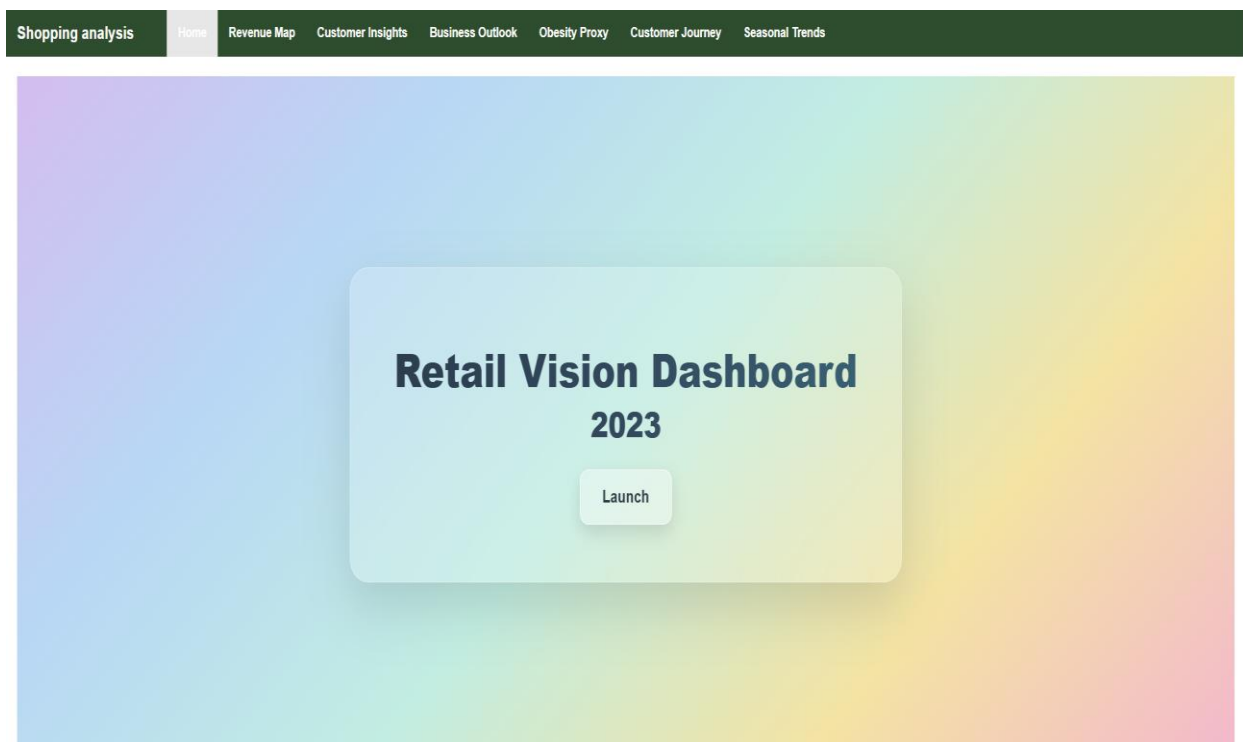


FIGURE: 1

The home page was designed to give the dashboard a website-like feel, with a navigation bar across the top linking to six analytical tabs Revenue Map, Customer Insights, Business Outlook, Obesity Proxy, Customer Journey and Seasonal Trends. The Launch button in the centre takes users straight into the Revenue Map.

4.2) GEO-REVENUE LENS (CHOROPLETH MAP)



FIGURE: 2

The dashboard gives a geographic view of retail revenue across U.S. states. The main element is an interactive map where each state is outlined and labelled so that regional comparisons can be made easily. The top five revenue states are marked with star icons, and a legend on the right shows their ranking from first to fifth. A hover feature was also added to the map. Placing the cursor over any state brings up the state name and its revenue figure on screen.

One of the more technically complex parts of this project was combining spatial data with transaction revenue data. Getting the map built properly started with bringing in official U.S. state boundary files through the tigris package. The sf package was then used to work with the spatial data. Reprojecting the coordinate system was a necessary step before the polygon geometries could be joined up with the revenue figures. With that done, Leaflet handled the rendering of the final interactive map. **[Question d]**

Users can zoom in on specific regions using the zoom controls. The side panel lets users choose a product category, switch between seasonal and annual views, and render between seasons using a slider. A summary table below the map lists the top states with their revenue figures. Both green and blue colours carry geographic associations, which made them a reasonable fit for this type of spatial analysis. The base-map was kept underneath the fills so geographic context was not stripped away entirely. Each state got a different colour to prevent neighbouring boundaries from blending together.

4.3) CUSTOMER DEMOGRAPHICS (RIDGELINE PLOT)

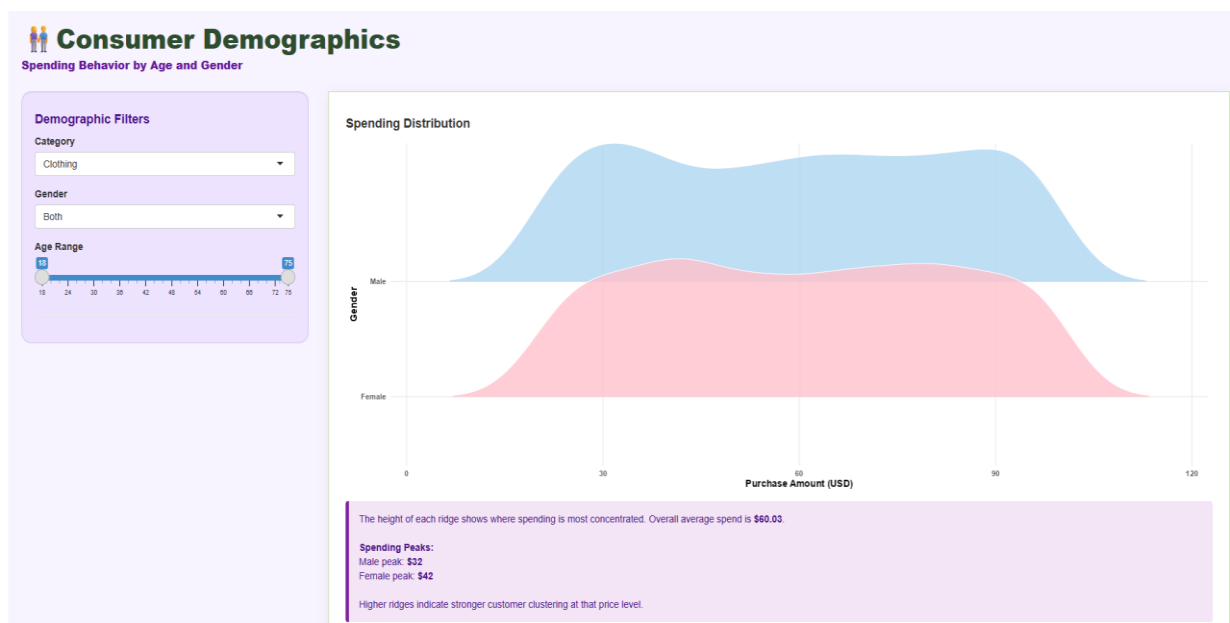


FIGURE: 3

This dashboard looks at spending behaviour broken down by age and gender for a selected product category. The horizontal axis shows purchase amounts, and separate density curves are drawn for male and female customers. The shape of each curve indicates where spending is most concentrated for that particular group. A sharper and taller peak at a certain point on the axis suggests that a greater number of customers from that group spent close to that amount. Average spending and peak values are also shown in a summary section for extra clarity.

Pink was chosen for female customers while blue was used for male customers, largely because both colours are already widely understood as gender indicators and required no further clarification within the chart. The background colour selected was a light purple. This came from the idea that purple is essentially what you get when pink and blue are combined, and since the chart was built around comparing two gender groups together, that connection felt worth making visually.

4.4) REVENUE OVER TIME (TIME-SERIES PLOT)

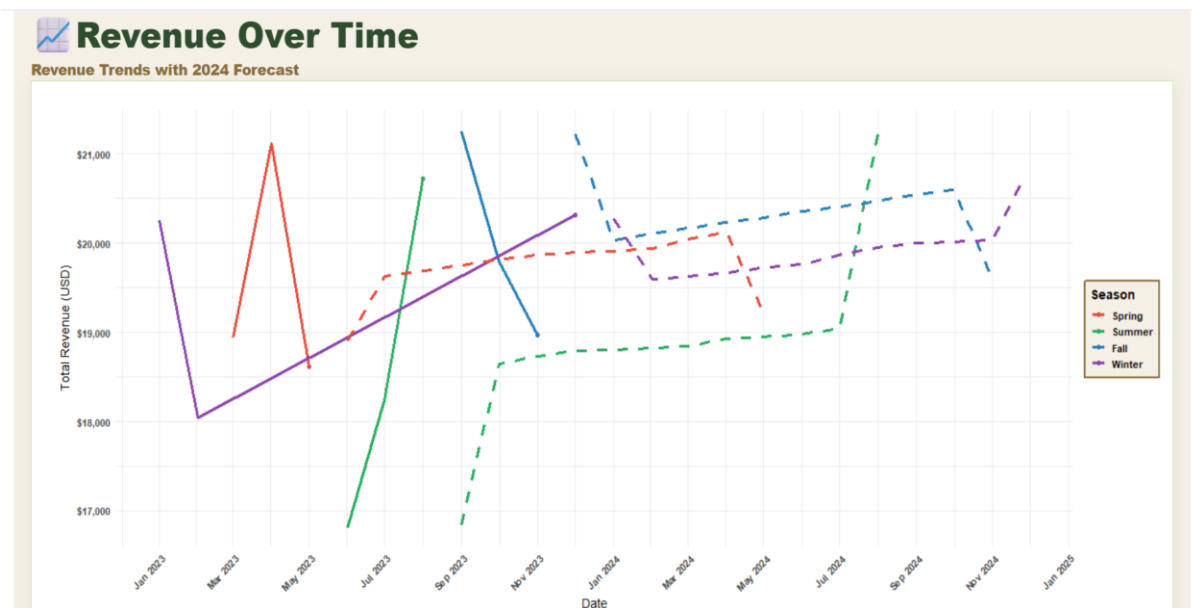


FIGURE: 4

This dashboard was put together to show how revenue shifted over time, with a forecast element added covering 2024. The vertical axis carries total revenue in USD while time moves along the horizontal. Keeping each season as its own separate line worked well in practice, as revenue dips and growth periods that would have blended together on a single line became far easier to identify this way. The forecast values are included to give a forward-looking view. Giving each season its own distinct colour meant individual trends could be followed throughout the chart without any confusion between them.

(Yau, 2011) made the point that visualisation brings out patterns and trends in data that raw numbers on their own rarely make obvious. The revenue line chart is a fair example of this in practice, as seasonal shifts and the 2024 forecast are both things that a table of numbers would struggle to communicate as clearly.

4.5) OBESITY PROXY RANKING (LOLLIPOP CHART)

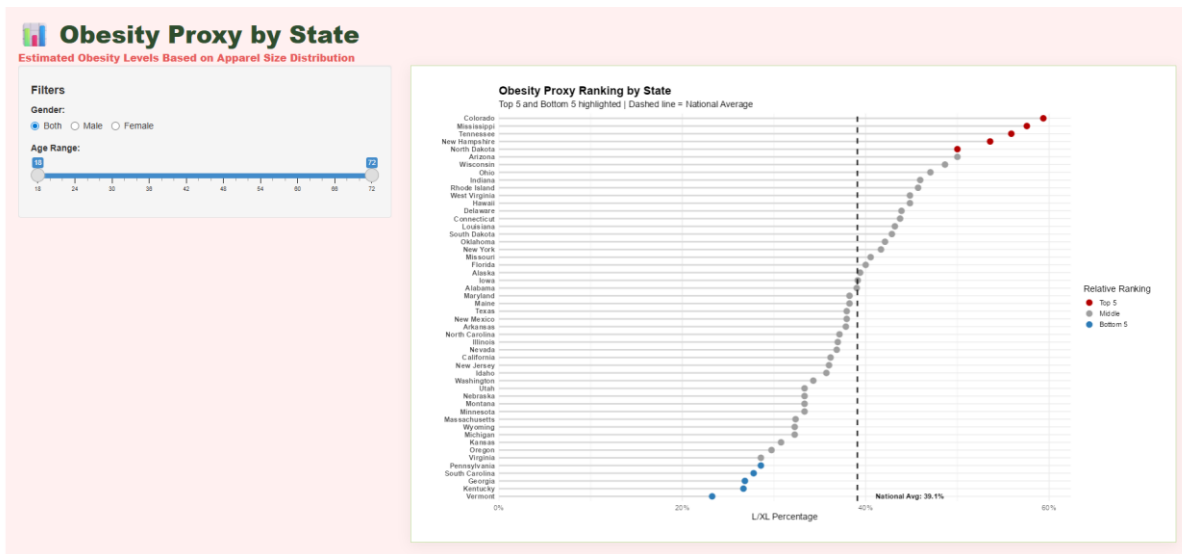


FIGURE: 5

This chart shows the percentage of L/XL apparel purchases per state. States run down the vertical axis in ranked order. The horizontal axis tells you the L/XL purchase percentage for each one. A dashed vertical line marks the national average and this made it straightforward to see which states fell above or below that figure.

The colour coding is based on where each state sits in the distribution. Red was used for the five states sitting at the top of the ranking. These states had the highest proportion of L/XL purchases, which could point toward higher obesity proxy levels in those areas. The bottom five states were coloured blue and those sitting close to the national average were left in grey. Gender and age range filters on the left side of the panel let users look at specific groups within the ranking.

4.6) CUSTOMER PATHWAY ANALYSIS (SANKEY DIAGRAM)

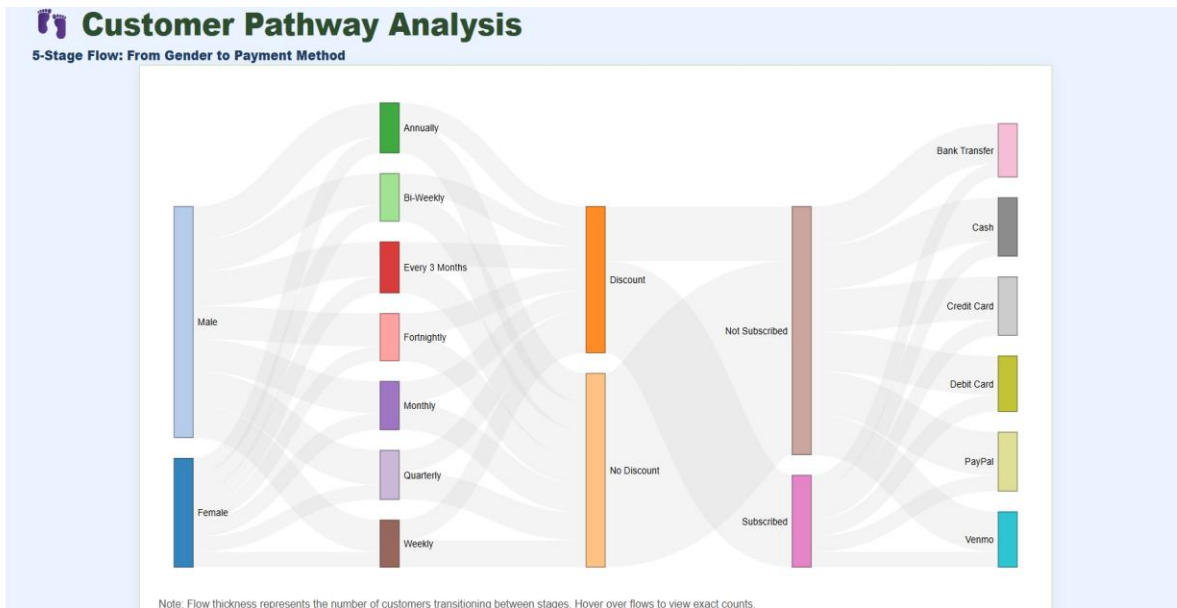


FIGURE: 6

This diagram follows customers across five stages, starting at gender and ending at payment method, with purchase frequency, discount usage and subscription status sitting in between. How thick each connecting flow is depends on how many customers took that particular path, which makes the more travelled routes easy to identify. Separate colours were used at each stage purely to make sure one variable did not get mixed up with another.

According to (Tufte, 2001), a well-designed visualisation should maximise the data-ink ratio, meaning most of the ink on the page should be there to represent actual data rather than decoration. This holds true for the Sankey diagram used here. Nearly everything drawn on the chart is there because it stands for a real customer movement between stages. Decorative or unnecessary elements were kept to a minimum. This makes the diagram a fairly efficient way of representing the data without overcomplicating it.

4.7) SEASONAL STYLE PULSE (BUBBLE CHART)

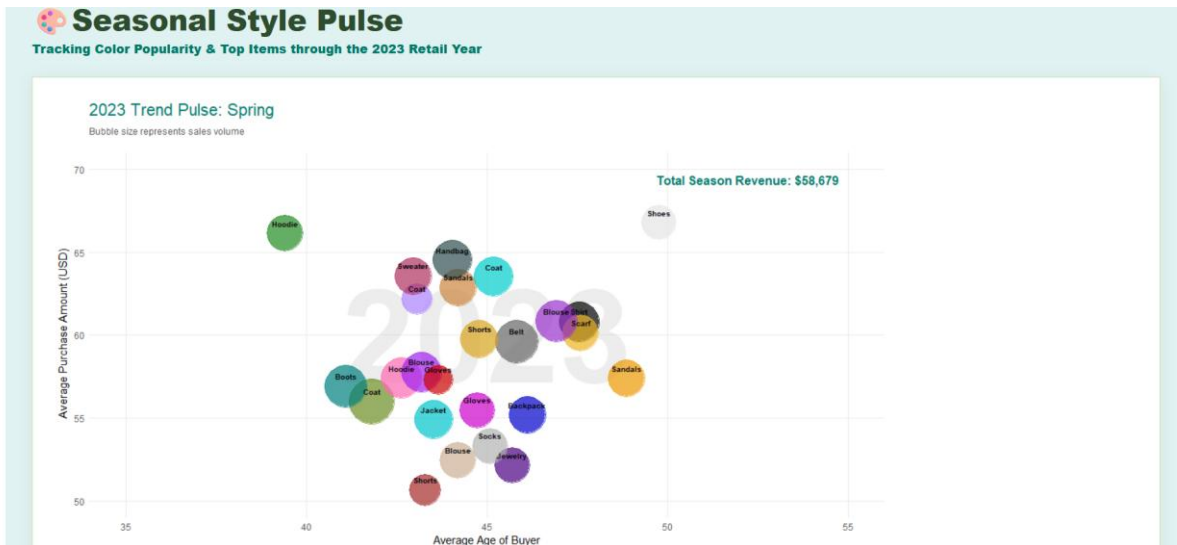


FIGURE: 7

This bubble chart looks at how products perform across the four seasons. The horizontal axis carries average buyer age, the vertical axis holds average purchase amount in USD, and how big each bubble is depending on sales volume. Putting all of this together in one view made spotting popular products and spending differences across seasons much more manageable.

(Tuft, 2001) suggested that, effective visualisations should display multiple variables without losing clarity in the process. Looking at the bubble chart, age, purchase amount, item type, season and volume are each communicated through position, size and colour, and despite carrying that much information the chart does not feel overwhelming. Bubble colours were taken directly from the actual item colours customers bought, connecting the visual to genuine purchasing behaviour. Total revenue is also shown in the chart to give an immediate sense of the financial performance of each product.

5) THEORETICAL AND DESIGN CONSIDERATIONS

[Question e]

Three main design decisions were made during the development of this project. The first was choosing the right chart type for each part of the data. Ridgeline density plots were used for the gender spending comparison as they give a clear picture of how each group's spending is distributed and where the two overlap, something a bar chart or line graph would not show as well. The Sankey diagram was a natural fit for the customer journey given how well it captures movement through sequential stages. Revenue data was mapped geographically since regional differences between states are something a map communicates far better than a table ever could. Every chart choice came back to the same question of what worked best for that specific part of the data.

The second decision was around how to use interactivity and animation. Filters were made interactive so users can explore the data on their own rather than just reading a static chart. Animations were added to make seasonal and time-based shifts easier to follow. The aim was to give users something to interact with rather than a static set of charts.

The third decision was around colour and visual hierarchy. Colour consistency was maintained across the dashboard while still keeping each section visually distinct from the others. Where contrast was applied such as on the national average line or top performing states, it was done to make sure the most important values caught the eye first, regardless of the viewer's background.

6) NOVELTY OF APPROACH

[Question c]

The novelty of this project stemmed from bringing multiple angles of analysis together in one place rather than treating each chart separately. Adding the synthesised Purchase Date column opened up the time-series and seasonal analysis that the original dataset did not support. The obesity proxy chart idea was also a completely original idea, it was not something found in any existing visualisation of this dataset. These two additions made this version of the analysis distinct from anything previously published.

The dashboard was built to be colourful and easy to get around, with the idea that the data complexity should never make things confusing for whoever is using it. Technical accuracy was one priority, but making the outputs readable for a non-specialist was just as much of one.

7) FUTURE IMPROVEMENTS

[Question g]

Given more time, a few areas could have been taken further. ARIMA or Machine Learning based forecasting could replace the current approach to give more accurate revenue projections. Bringing in validated demographic data would make the obesity proxy findings a lot more solid. The dashboard would also need some technical work to keep it running well once the data volume gets bigger.

8) CONCLUSION

This project demonstrated how interactive visualisation can reveal and communicate patterns in retail shopping behaviour. An R shiny dashboard was developed covering geographic, demographic, time-based and behavioural analysis in one place. Adding the synthesised purchase date column pushed the analysis beyond what the original data allowed. The priority throughout was keeping the visuals clear and the findings accessible to any audience.

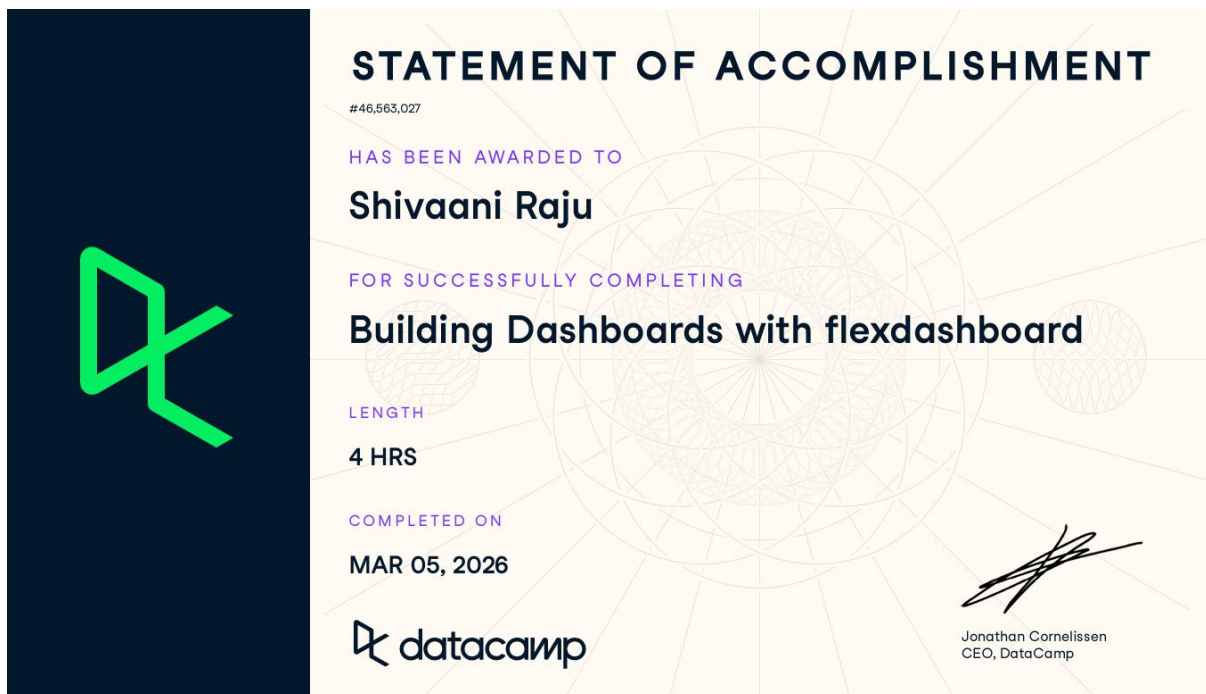
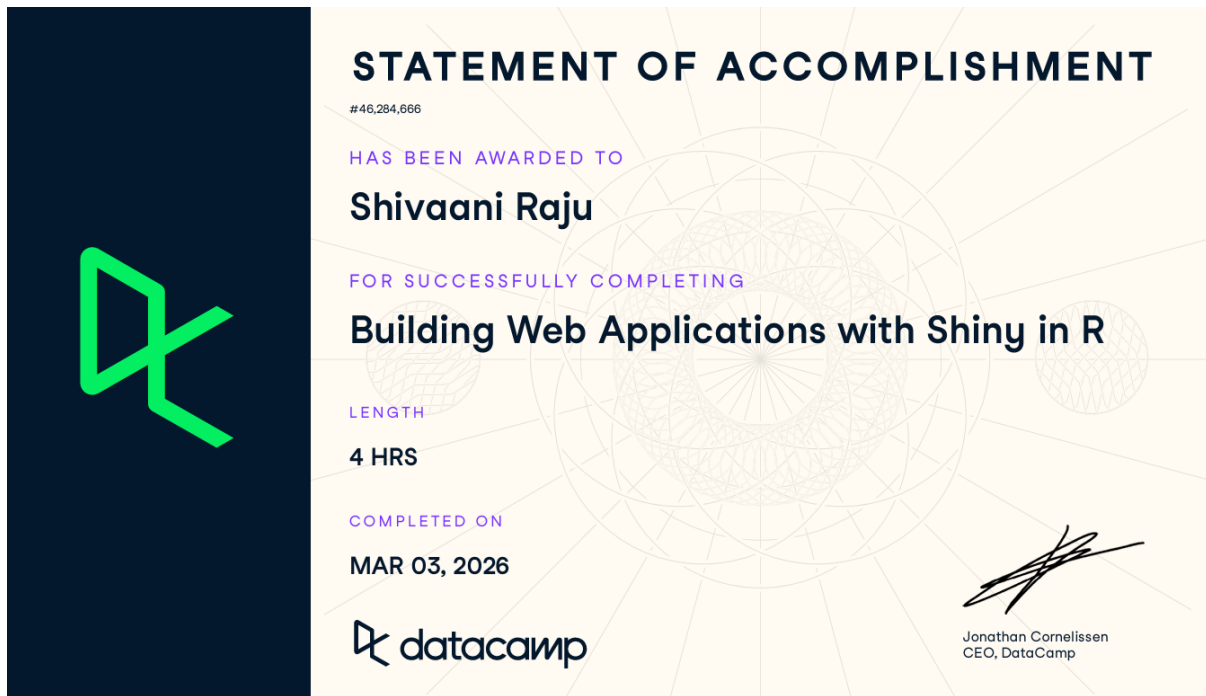
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DATACAMP CERTIFICATES



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